

Engineering Economics (ENCE 307) - Micro Syllabus

Course Information	Details
Year/Part	III / I
Teaching Schedule	L:3, T:1, P:0 (45 Total Lecture Hours)
Depth Codes Used	D -Definition, E -Explanation, NUM -Numerical, DV -Derivation, DW -Drawing, PS -Problem Solving, I -Illustration

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual Plan (L:T:P)	Week
1	Introduction (2 Hours)			2:1:0	1
	1.1 Micro, macro and engineering economics	D, E	History, fundamental principles, and application of economics in engineering.		
	1.2 Terminology & Decision Making	D, E	Economic terminology, the role of engineers in decision making.		
	1.3 Cash flow and Cash Flow Diagram (CFD)	D, DW, PS	Concept of Cash Flow, Construction of Cash Flow Diagrams (CFD).		
2	Market Economics (3 Hours)			3:1:0	1-2
	2.1 Market, demand, supply and relationship	D, E, I	Relationship between demand and supply, and Market equilibrium.		
	2.2 Elasticity and government policies	NUM, PS	Application of elasticity, government policies (e.g., price ceilings/floors).		
	2.3 Market failures & Firm behavior	E, D	Externality, market inefficiency, market failure, and firm behavior.		
3	Cost (2 Hours)			2:1:0	2
	3.1 Cost Classification	D, E	Total, average, fixed, variable, marginal, direct, indirect, standard, sunk, opportunity, and life-cycle costs.		
	3.2 Cost estimation and control	E, NUM	Cost estimation techniques and cost control.		
4	Time Value of Money (6 Hours)			6:1:0	3-4
	4.1 Money & Interest	D, NUM	Functions of money, Time value of money, Simple vs Compound interest.		
	4.2 Compounding	NUM, PS	Nominal, effective, and continuous compounding calculations.		

	4.3 Equivalence & Cash Flows	NUM, DV	Economic equivalence, analysis of Single, Uniform, Gradient (Linear & Geometric), and Irregular cash flows.		
5	Methods of Economic Analysis (12 Hours)			12:2:0	5-8
	5.1 Capital Budgeting & MARR	D, E	Concept of Capital Budgeting and Minimum Attractive Rate of Return (MARR).		
	5.2 Equivalent Worth Methods	NUM, PS	Application of Net Present Worth (NPW), Annual Equivalent Worth (AEW), Future Worth (FW), Capitalized Cost (CW).		
	5.3 Rate of Return	NUM, PS	Calculation of Internal Rate of Return (IRR) and External Rate of Return (ERR).		
	5.4 Public Sector & WACC	D, NUM, PS	Benefit-Cost Analysis (B/C Ratio) and Weighted Average Cost of Capital (WACC).		
	5.5 Project Comparisons	E, PS	Analysis of Independent, Dependent, and Contingent projects; Repeatability assumptions.		
6	Replacement Analysis (5 Hours)			5:1:0	9-10
	6.1 Replacement Strategies	D, E	Concept of Defender vs. Challenger.		
	6.2 Economic Service Life (ESL)	NUM, PS	Calculation of the Economic Service Life (ESL) of an asset.		
	6.3 Horizon Analysis	E, PS	Replacement strategies for finite and infinite planning horizons.		
7	Risk Analysis (5 Hours)			5:1:0	10-11
	7.1 Risk Analysis Techniques	NUM, PS	Sensitivity analysis, Breakeven analysis, and Scenario analysis.		
	7.2 Decision Tree	DW, PS	Construction and analysis of Decision Trees for decision making under risk.		
8	Depreciation and Taxes (5 Hours)			5:1:0	12-13
	8.1 Depreciation Methods	D, NUM	Straight line, Declining balance, Sum of Years' Digits (SOYD), and MACRS.		
	8.2 Taxes & After-Tax Analysis	NUM, PS	Corporate income tax, Calculation of After-Tax Cash Flow (ATCF).		
9	Measurement of National Income (5 Hours)			5:1:0	14-15

	9.1 GDP & Unemployment	D, E	Real vs. Nominal Gross Domestic Product (GDP), measurement of unemployment.		
	9.2 Inflation	E, NUM	Causes/effects of inflation, and equivalence calculations under inflation.		
	9.3 Exchange & Fiscal Policy	E, S	Real vs. Nominal exchange rates, Fiscal budget, and Monetary policy.		

Model Question

QN	Question	Marks	Unit
1	Explain the concept of Economic Equivalence and Time Value of Money (TVM) . Derive the formula for the Present Worth (P) of a Uniform Series (A) for n periods at an interest rate i . [DV]	10	4
2	What is Depreciation ? Explain the difference between book value and market value . A heavy machine costing Rs. 50,00,000 has a useful life of 8 years and a salvage value of Rs. 2,00,000. Calculate the depreciation charge and the book value for the 4th year using both the Straight-Line Method and the Sum-of-Years'-Digits (SOYD) Method . [D, NUM, PS]	10	8
3	Discuss the various types of cost classification used in engineering project evaluation. Explain the significance of Sunk Cost and Opportunity Cost in decision-making. Illustrate the process of After-Tax Cash Flow (ATCF) calculation for a project, clearly indicating how depreciation expense serves as a tax shield . [D, E, I, PS]	10	3, 8

QN	Question	Marks	Unit
4	Define elasticity of demand and explain its practical application in setting prices for a public utility service (e.g., electricity or water supply). Discuss the impact of government policies like price ceilings on market equilibrium. [D, E, I]	5	2

	Distinguish between Internal Rate of Return (IRR) and External Rate of Return (ERR) . Why is ERR sometimes considered a more reliable measure than IRR for project selection? [D, E]	5	5
6	A company is considering two mutually exclusive projects, A and B, both having a life of 5 years. Given a MARR of 10%, use the Net Present Worth (NPW) Method to select the better project. [NUM, PS] Project A: Initial Cost = Rs. 10,00,000, Annual Benefit = Rs. 3,50,000. Project B: Initial Cost = Rs. 8,00,000, Annual Benefit = Rs. 3,00,000.	5	5
7	Explain the concept of Capitalized Cost (CW) . A dam requires an initial investment of Rs. 5 Crore and major maintenance of Rs. 50 Lakh every 10 years. If the interest rate is 8%, calculate the capitalized cost of the dam. [D, NUM, PS]	5	5
8	Discuss the major causes and effects of inflation on an economy. How are real and nominal interest rates related in an inflationary environment? Provide an example of how inflation affects equivalence calculations. [E, NUM]	5	9
9	What are the steps involved in a Replacement Analysis ? Determine the Economic Service Life (ESL) for an asset with the following costs at an 8% interest rate: Initial Cost = Rs. 10,00,000, Annual Operating Cost: Year 1= Rs. 50,000, Year 2= Rs. 60,000, Year 3= Rs. 75,000. [E, NUM, PS]	5	6
10	Explain Sensitivity Analysis and Breakeven Analysis as tools for risk evaluation in engineering projects. How does a Decision Tree help a decision-maker under conditions of risk and uncertainty? [E, I, DW]	5	7
11	Differentiate between Microeconomics and Macroeconomics . Briefly explain the components used in the calculation of Gross Domestic Product (GDP) . [D, E]	5	1, 9

12	Distinguish between Weighted Average Cost of Capital (WACC) and Minimum Attractive Rate of Return (MARR). Under what circumstances would a project be considered acceptable based on the B/C Ratio method? [D, E]		
----	---	--	--